

NISHANT CHANDNA

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EDUCATION

Delhi Technological University

August 2023 – Present

B.Tech in Mechanical Engineering, Specialization in Automotive Engineering

CGPA: 9.42

HONORS AND AWARDS

- **2nd Place (Self-Funded Category)**, **DARPA Triage Challenge** Phase 2 (\$150,000 prize) and Phase 1 (\$60,000 prize).
- **2nd Place** in the simulation stage and **3rd Place** in the physical stage, **ICUAS 2024 UAV Competition**.
- **Global Innovation Excellence Award** (2025 and 2026), W-AHEAD Foundation.
- **Research Excellence Award**, Delhi Technological University.

EXPERIENCE

Founding Research Intern

May 2026 – Present

FPV Labs, Bangalore

Egocentric SLAM · sim-to-real manipulation

- Validated phone-based ARKit tracking against a Vicon motion-capture system across a range of motion types, to confirm it gives accurate poses at true metric scale.
- Wrote a Python evaluation suite that measures absolute and relative trajectory error, drift over time, and consistency when revisiting a place, and used it to compare classical SLAM (ORB-SLAM3, VINS-Fusion, OKVIS2) with learned SLAM (DROID-SLAM, MAST3R-SLAM, VGGT-SLAM).
- Built **LiV**, a LiDAR-grounded real-to-sim pipeline: a phone rigidly coupled to a 360° Livox Mid-360 turns a handheld capture into photorealistic, sim-ready USD assets. The LiDAR supplies every metric quantity (scale, gravity, splat depth, object size), a VLM infers mass and material, and each attribute is tagged measured, derived, human-confirmed or inferred.
- Used MimicGen-style data generation in that scene to expand a handful of teleoperated SO-101 demonstrations into a large synthetic dataset, keeping only episodes that complete the task.

Volunteer Researcher

April 2026 – Present

SAIR Lab, University at Buffalo

Vision-language navigation

- Integrating the lab's vision-language navigation stack with **OmniNavBench** (RSS 2026), a multi-embodiment navigation benchmark. Wrote the policy-server bridge that feeds the policy ground-truth odometry and depth-projected scans, bypassing the benchmark's intrinsics- and TF-dependent depth interface.

Research Intern

Feb 2026 – May 2026

RRC, IIIT Hyderabad (Prof. Sourav Garg)

Learned planning costs for navigation

- Developed **Plann3r**, which predicts a per-pixel geodesic cost map from a query image to a subgoal pixel in a submap of reference images. Built on VGGT: a learnable goal token added to one patch token marks the subgoal, and an MLP head maps each patch embedding to a cost.
- Fine-tuned VGGT's global-attention layers and the cost head with a log-scale Huber cost loss, a multi-scale Sobel gradient loss to keep obstacle boundaries sharp, and an auxiliary pointmap loss to preserve the backbone's geometry.

Research Intern

May 2025 – July 2025

RRC, IIIT Hyderabad (Prof. Madhava Krishna)

LiDAR odometry and mapping

- Added degeneracy detection and localizability analysis to LiDAR odometry and mapping, so the estimator handles scenes where some motion directions are poorly constrained.
- Added Scan Context place recognition to detect revisited places and a factor graph to apply the resulting loop closures, so drift accumulated by ICP odometry is corrected across the whole map.

Vice Captain & Autonomy Lead

August 2023 – Present

UAS-DTU, Delhi Technological University

Multi-robot autonomy

- As Vice Captain, lead a team of 30 undergraduate students competing in the DARPA Triage Challenge.
- Head the autonomy sub-team, responsible for navigation and exploration in unknown environments.
- Developed and integrated perception, planning and control modules for a team of ground and aerial robots.

PUBLICATIONS

Plann3r: Predicting Planning Costs Grounded in 3D

2026

Conference on Robot Learning (CoRL) 2026

- Proposed VGGT-Nav, a single-camera navigation pipeline built on Plann3r. Offline, costs are propagated back from the goal through overlapping submaps to give every reference image a global cost map. Online, Plann3r localizes the query through cross-frame attention, without explicit feature matching, and its cost map conditions a learned waypoint controller.
- Because the costs are geodesic rather than straight-line, the robot can reason about walls and doorways: it reaches goals that were seen but never visited on the prior route and takes shortcuts the demonstration did not. On these tasks it clearly outperforms image-level (GNM, PixNav) and object-level (RoboHop, ObjectReact) methods in HM3D.

DAMM-LOAM: Degeneracy-Aware Multi-Metric LiDAR Odometry and Mapping

2025

IIROS 2025 Workshop on Active Perception

- LiDAR odometry and mapping method that stays accurate in degenerate scenes such as long featureless corridors, where the geometry leaves some directions of motion poorly constrained.

- Outperforms existing LOAM methods in both degenerate and outdoor scenes.

Autonomous UAV Navigation and Mapping for Accurate Fruit Detection and Counting

2025

International Conference on Unmanned Aircraft Systems (ICUAS)

- UAV fruit counting with YOLOv8 detection, LiDAR-camera fusion for 3D mapping, and TSP with A* path planning.
- Fruits are localized in 3D so each one is counted once even when seen from several viewpoints; the system placed 2nd in simulation and 3rd in the physical stage of the ICUAS 2024 UAV Competition.

PROJECTS

FairEval | *LeRobot, PyTorch, OpenCV*

- Built an evaluation harness for fair comparison of real-robot policy checkpoints: every checkpoint runs the same counterbalanced object layouts in the same order, restaged against reference frames captured in the first run, so layout and framing differences are not attributed to the checkpoint.
- Measured the robustness of a task-finetuned VLA policy to camera intrinsics (focal length up to $2\times$), defocus, lighting and framing over a 161-episode, 129k-frame dataset, re-encoding the baseline through the same video pipeline so compression stays constant across conditions.

Drone-Based Forest Mapping | *Python, Bash, PX4, ROS, Git*

- Deployed a PX4-based UAV in a simulated Gazebo forest; implemented forest mapping with GenZ-ICP and benchmarked it against KISS-ICP; fused ICP odometry with GPS and IMU data using confidence-based dynamic weighting for state estimation.
- Designed custom simulation environments to evaluate the mapping and localization pipeline.

Collision Avoidance using EGO-Planner and VINS-Mono | *Python, C++, VIO, PX4, ROS*

- Implemented GPS-denied autonomous navigation and collision avoidance indoors, using VINS-Mono for visual-inertial state estimation and EGO-Planner for real-time local trajectory generation.
- Integrated perception, planning and control on a lightweight quadrotor running PX4 and ROS for fast flight through cluttered indoor spaces.

PyFlyt: Reward Shaping for Drone RL (Open Source) | *Python, PyBullet, Gymnasium, Stable Baselines 3*

- Contributed a quadratic yaw-rate penalty to the open-source PyFlyt drone RL library across its QuadX hover, waypoint, pole-balancing and ball-in-cup environments, curbing aggressive yawing to improve flight stability; merged upstream and released in PyFlyt 0.29.0.

TECHNICAL SKILLS

Robotics & Simulation: ROS/ROS 2, PX4, Gazebo, Isaac Sim, Isaac Lab, OmniGibson, ArduPilot SITL, LeRobot, Docker, Git
Perception & Estimation: SLAM, VIO/LIO, Sensor Fusion, Factor Graph Optimization (GTSAM), Kalibr, Gaussian Splatting, 3D Reconstruction

Learning: Python, PyTorch, Imitation Learning, Vision-Language Navigation, VLA Policies, Reinforcement Learning (Stable Baselines 3)

Hardware: Jetson Nano, Jetson Orin, Pixhawk, Arduino, LiDAR, Intel RealSense, OAK-D, SO-101 Arm